

# Overfitted high-dimensional matrix factorizations via adaptive spectral shrinkage

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# Main Contribution

We introduce EigenBayes, a scalable method for **high-dimensional factor analysis**, which combines

- 1 **spectral estimators,**
- 2 **adaptive Bayesian shrinkage.**

## Motivation: Covariance estimation in high dimensions

- Scientific datasets often contain many more variables than samples: gene expression, imaging, multi-omics, environmental exposures, etc.
- A full covariance matrix has  $O(p^2)$  parameters, where  $p$  is the data dimension.
- Factor models reduce the problem to a low-rank signal plus residual noise.

## Factor model setup

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- $\Sigma = \text{diag}(\sigma_1^2, \dots, \sigma_p^2)$ : variable-specific residual noise.

# Covariance induced by the factor model

Starting from

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## Matrix representation

$$Y = M \Lambda^\top + E, \quad Y = \begin{bmatrix} y_1^\top \\ \vdots \\ y_n^\top \end{bmatrix}, \quad M = \begin{bmatrix} \eta_1^\top \\ \vdots \\ \eta_n^\top \end{bmatrix}, \quad E = \begin{bmatrix} \epsilon_1^\top \\ \vdots \\ \epsilon_n^\top \end{bmatrix}.$$

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Given a latent dimension  $k$ , the standard spectral estimators (e.g. Bai [1]) use the leading  $k$  components of the SVD of  $Y$

$$\hat{M} = \sqrt{n} U_{1:k}, \quad \hat{\Lambda}_{\text{svd}} = \frac{1}{\sqrt{n}} V_{1:k} D_{1:k}.$$



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### Limitation

The estimator depends critically on the choice of  $k$ . Weak but real components may be missed, while overestimating  $k$  introduces noisy directions.

## Bayesian overfitted models: robust but computationally expensive

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## EigenBayes

Retain scalability and simplicity of **spectral estimates** together with **adaptivity** of Bayesian methods.



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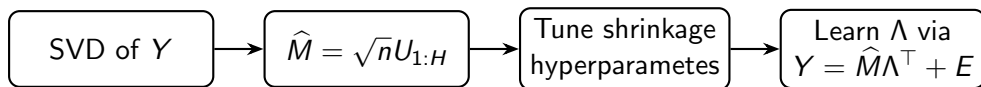
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## Step 1: Overfitted spectral estimator for the factors

Recall the SVD of  $Y$

$$Y = UDV^{\top},$$

### Initial factor estimate

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- If  $H = k$ , this is the usual spectral factor estimator.
- If  $H > k$ , extra directions mostly represent noise.
- EigenBayes keeps those extra directions, but shrinks them adaptively.

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Condition on  $\hat{M}$  and rewrite the model as

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For each outcome  $j = 1, \dots, p$ ,

$$y^{(j)} = \hat{M}\tilde{\lambda}_j + \tilde{e}^{(j)}, \quad \tilde{e}_i \sim N_p(0, \tilde{\Sigma}).$$

where  $y^{(j)}, \tilde{e}^{(j)}$  are the  $j$ -th column of  $Y$  and  $\tilde{E}$  and  $\tilde{\lambda}_j$  be the  $j$ -th row of  $\tilde{\Lambda}$ :

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- $\tau_j^2$  controls outcome-specific signal-to-noise ratio.
- $\Psi = \text{diag}(\psi_1^2, \dots, \psi_H^2)$  controls component-specific shrinkage.

# Conjugate posterior update

This formulation admits a conjugate update.

## Posterior mean

The posterior mean for  $\tilde{\lambda}_j$  is

$$\hat{\lambda}_j = \sqrt{n} \Psi_{n,j}^{-1} D_{1:H} v_{j,1:H}, \quad \Psi_{n,j} = \text{diag} \left( n + \tau_j^{-2} \psi_1^{-2}, \dots, n + \tau_j^{-2} \psi_H^{-2} \right).$$

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- $\sqrt{n} \Psi_{n,j}^{-1}$ : outcome- and component-specific shrinkage factors.

## Connection to spectral shrinkage

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Therefore, component-wise,

$$\hat{\lambda}_{j\ell} = \underbrace{\frac{n}{n + \tau_j^{-2} \psi_\ell^{-2}}}_{\text{adaptive shrinkage}} \hat{\lambda}_{\text{svd},j\ell}, \quad \ell = 1, \dots, H.$$

## Step 3: Adaptive shrinkage

Recall the prior

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- $\psi_\ell$  controls the **a priori variance** (and expected squared magnitude) of the  $\ell$ -th latent component, i.e. the  $\ell$ -th column of  $\tilde{\Lambda}$  (and similarly the  $\tau_j^2$ 's with the rows).

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- **Rationale:**
  - ▶ Tune  $\{\psi_\ell\}_{\ell=1}^H$  and  $\{\tau_j^2\}_{j=1}^p$  so that

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- ▶ Because  $H$  is overfitted, the spectral magnitudes are corrected by subtracting the noise edge:

$$d_\ell^2 \rightsquigarrow d_\ell^2 - d_{H+1}^2,$$

as a proxy for of the  $\ell$ -th component signal's strength, where  $d_l$  and  $d_{H+1}$  are the  $l$ -th and  $H+1$ -th singular values of  $Y$ .

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### Closed-form hyperparameters

$$\hat{\psi}_\ell^2 = \frac{d_\ell^2 - d_{H+1}^2}{\sum_{r=1}^H (d_r^2 - d_{H+1}^2)} H, \quad \ell = 1, \dots, H,$$

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- Large signal direction  $\Rightarrow$  weak shrinkage.
- Superfluous direction  $\Rightarrow$  aggressive shrinkage.

# Why do we subtract $d_{H+1}^2$ ?

## Without correction

Matching the prior expectation directly to the spectral estimate would include noise from overfitted components:

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## Noise-corrected moment matching

Instead use

$$\hat{\psi}_\ell^2 = H \frac{d_\ell^2 - d_{H+1}^2}{\sum_{m=1}^H (d_m^2 - d_{H+1}^2)}.$$

$$\ell \leq k : d_\ell^2 \gg d_{H+1}^2 \Rightarrow \hat{\psi}_\ell^2 \text{ remains large,}$$

$$\ell > k : d_\ell^2 \approx d_{H+1}^2 \Rightarrow \hat{\psi}_\ell^2 \approx 0.$$



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## Without correction

Matching the prior expectation directly to the spectral estimate would include noise from overfitted components:

$$d_\ell^2 \rightsquigarrow \hat{\psi}_\ell^2 \propto d_\ell^2.$$

Then superfluous components may retain non-negligible prior variance, so performance can deteriorate as  $H$  increases.

## Noise-corrected moment matching

Instead use

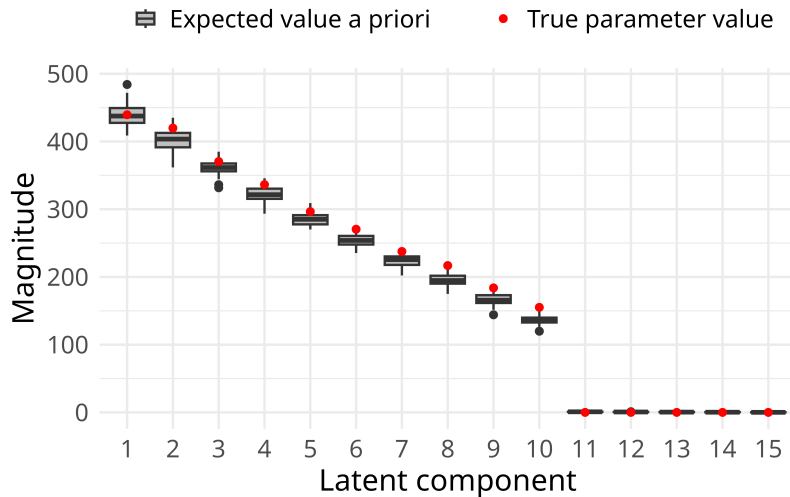
$$\hat{\psi}_\ell^2 = H \frac{d_\ell^2 - d_{H+1}^2}{\sum_{m=1}^H (d_m^2 - d_{H+1}^2)}.$$

$$\ell \leq k : d_\ell^2 \gg d_{H+1}^2 \Rightarrow \hat{\psi}_\ell^2 \text{ remains large,}$$

$$\ell > k : d_\ell^2 \approx d_{H+1}^2 \Rightarrow \hat{\psi}_\ell^2 \approx 0.$$

**This is what allows EigenBayes to automatically shrink superfluous latent components.**

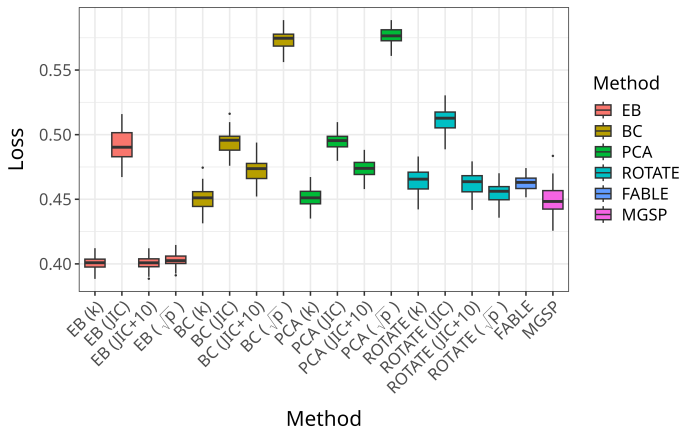
## Adaptive shrinkage in practice ( $k = 10$ vs $H = 15$ )



## Numerical experiments – setup

- Compare EigenBayes with BC (Barigozzi and Cho [2]), PC (Bai [1]), ROTATE (Rockova and George [5]), FABLE (Chattopadhyay et al. [4]), and MGSP (Bhattacharya and Dunson [3]).
- Generate data from Gaussian factor models and loadings generated under our prior with  $k = 10$ ,  $n = 500$ ,  $p = 5000$  (other scenarios in the paper).
- Loss: relative Frobenius error for estimating  $\Theta$ .

## Simulation results: overfitting the latent dimension

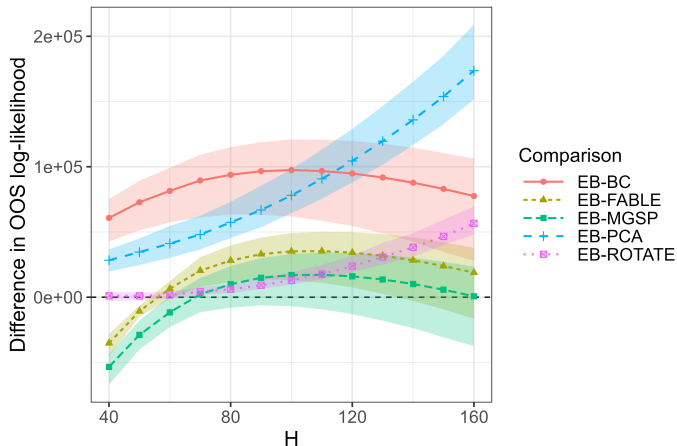


### Takeaway

EigenBayes remains accurate when  $H$  is overfitted, while several spectral competitors deteriorate as the selected rank increases.

## Application: ImmGen gene expression

- ImmGen microarray dataset with  $n = 628$  samples and  $p = 5000$  high-variance genes.
- Evaluate out-of-sample log-likelihood under estimated covariance over 50 random 80/20 train-test splits.



# Conclusion

- EigenBayes is a scalable method for overfitted factor models.
- Adaptive shrinkage acts across both latent components and outcomes.
- Superfluous components are shrunk toward zero, reducing sensitivity to rank selection.
- The method provides covariance estimation, latent factor inference, and valid uncertainty quantification without MCMC.

# Thank you!

Questions?

**arXiv:** ML. and Dunson, D. B. (2026+).

*Overfitted high-dimensional matrix factorizations via adaptive spectral shrinkage.*

arXiv:2606.19540



# References I

- [1] Jushan Bai. “Inferential Theory for Factor Models of Large Dimensions”. In: *Econometrica* 71.1 (2003), pp. 135–171.
- [2] Matteo Barigozzi and Haeran Cho. “Consistent estimation of high-dimensional factor models when the factor number is over-estimated”. In: *Electronic Journal of Statistics* (2018).
- [3] A. Bhattacharya and D. B. Dunson. “Sparse Bayesian infinite factor models”. In: *Biometrika* 98.2 (June 2011), pp. 291–306. ISSN: 0006-3444. DOI: 10.1093/biomet/asr013.
- [4] Shounak Chattopadhyay, Anru R Zhang, and David B Dunson. “Blessing of dimension in Bayesian inference on covariance matrices”. In: *arXiv preprint arXiv:2404.03805* (2024).
- [5] Veronika Rockova and Edward I. George. “Fast Bayesian Factor Analysis via Automatic Rotations to Sparsity”. In: *Journal of the American Statistical Association* 111.516 (2016), pp. 1608–1622.



## Formula for the row shrinkage parameters

For each outcome  $j = 1, \dots, p$ ,

$$\hat{\tau}_j^2 = \frac{v_{j,1:H}^\top (D_{1:H}^2 - d_{H+1}^2 I_H) v_{j,1:H}}{nH\hat{\sigma}_j^2}.$$

- Numerator: noise-corrected spectral signal assigned to outcome  $j$ .
- Denominator: residual variance estimate times normalization  $nH$ .
- Large  $\hat{\tau}_j^2$ : weak row shrinkage; small  $\hat{\tau}_j^2$ : strong row shrinkage.

# Theoretical guarantees

Assume  $p/n \rightarrow c \in (0, \infty)$ ,  $H = \mathcal{O}(n^{2/5})$ , and some other regularity conditions:

- **Automatic rank adaptation:** for  $\ell > k$ ,

$$\max_{j=1,\dots,p} \widehat{\psi}_\ell^2 \widehat{\tau}_j^2 \lesssim n^{-(1+2/5)}, \quad \widehat{\lambda}_j^{(\ell)} \xrightarrow{\text{pr}} 0.$$

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- **Posterior contraction:**

$$\frac{\|\hat{L} - L_0\|}{\|L_0\|} \lesssim \frac{\sqrt{\log n}}{\sqrt{n}}, \quad \frac{\|\hat{\Theta} - \Theta_0\|}{\|\Theta_0\|} \lesssim \frac{\sqrt{\log n}}{\sqrt{n}}.$$

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- **Entrywise inference:** Point estimates for the covariance entries satisfy a CLT and a Bernstein–von Mises result for the induced posterior.